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CRUCIBLE-G3 Rendering Subsystem

Sufeng Guo (sufeng_guo@smail.nju.edu.cn) · Microsoft Research Asia · Technical Design and Verification Report

Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 31 October 2026 (2026-10-31 23:59 UTC+8 Beijing Time)

Abstract

CRUCIBLE-G3 is a synthesizable fixed-point rendering subsystem preserving G1/G2 transformation, clipping, rasterization, depth testing and six-lane attribute interpolation. It adds a four-line 4x4 bilinear texture cache, programmable texture combination and alpha blending, synchronous RGBA8 framebuffer storage and tile-ordered DMA export. Software-vs-Hardware numerical comparison passed against independent integer models using simulated RTL across 15 configuration/seed combinations. L1 RTL timing lint passed across 22 files. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA achieved WNS_setup = +1.362 ns and hold slack = +0.044 ns at 10.0 ns for the standalone default-parameter implementation, with virtual interfaces and external I/O paths excluded. This establishes internal clocked timing, not board-level or integrated PCIe-bridge timing. L3 physical Arria 10 FPGA HIL passed three deterministic bridge-connectivity probes, not on-board rendering-vector comparisons. [E041, E046, E047, E048]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

crucible_g3_top serializes commands around one compatibility renderer and shared depth store. G3 draws remove only bit9 before replay through crucible_g2_top, retaining compatibility.legacy and compatibility.engine. Accepted fragments are optionally textured, shaded against a synchronously captured destination, written to color storage and serialized. Single-in-flight arithmetic and single-outstanding memory channels favor bounded logic depth and explicit recovery over throughput. Standalone parent copies use registered divider scheduling and limb-based interpolation accumulation without changing numerical behavior; the supplied parent reference remains unchanged. [E001, E002, E004, E005]

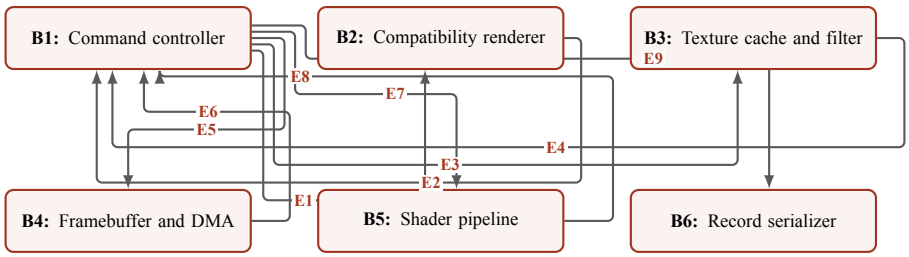


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Command controller	Buffers complete frames, validates descriptors and prerequisites, replays compatibility commands and sequences fragment processing.
B2: Compatibility renderer	crucible_g2_top retains legacy and engine children for geometry, rasterization, interpolation and shared depth storage.

Block	Responsibility
B3: Texture cache and filter	g3_texture_cache contains four tagged 4x4 lines and its g3_bilinear child, filter; gathers samples and performs acknowledged refills.
B4: Framebuffer and DMA	g3_framebuffer_dma owns synchronous RGBA8 storage, clear/read/write operations and tile-ordered export.
B5: Shader pipeline	g3_shader_pipeline quantizes colors, combines texture and applies independently programmed RGB/alpha blend factors.
B6: Record serializer	Preserves parent records and emits G3 fragment/completion words with stable stalled payloads.

Directed relationship	Meaning
E1: B1 to B2	Validated command replay
E2: B2 to B1	Fragments and terminal statistics
E3: B1 to B3	Configuration and texture coordinates
E4: B3 to B1	Filtered RGBA or error
E5: B1 to B4	Clear, read, write and export control
E6: B4 to B1	Destination color and completions
E7: B1 to B5	Attributes, texture, destination and program
E8: B5 to B1	Source or blended RGBA
E9: B1 to B6	Ordered record words

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Common clock and active-low asynchronous control reset. Every public output is zero during reset. One BOOT edge follows release; command transfer begins no earlier than the next edge.
s_valid, s_ready, s_data	Input valid/ data; output ready	1, 1, 32 bits	Command stream; pre-edge valid AND ready accepts exactly one word. Exact opcode decoding selects a 31- or 49-word frame.
m_valid, m_ready, m_ data, m_last	Output valid/ data/last; input ready	1, 1, 32, 1 bits	Ordered records; payload holds under backpressure. Only the final terminal word asserts last. Intake resumes after its handshake.
tex_req_valid, tex_req_ ready, tex_req_addr	Output valid/ address; input ready	1, 1, 32 bits	Unsigned byte-addressed texture reads, one outstanding request; address holds while stalled.
tex_rsp_valid, tex_rsp_ ready, tex_rsp_data, tex_ rsp_error	Input valid/ data/error; output ready	1, 1, 32, 1 bits	Response cannot precede the clock following request acceptance. Arbitrary fair delays are permitted; unsolicited responses are not.
dma_req_valid, dma_ req_ready, dma_req_ addr, dma_req_data, dma_req_strb, dma_req_ last	Output valid/ address/data/ strobes/last; input ready	1, 1, 32, 32, 4, 1 bits	Single-outstanding framebuffer writes; strobes equal 0xf when valid. Last marks only the final export request. All payload fields hold while stalled.

Signal / group	Direction	Width	Contract
dma_rsp_valid, dma_ rsp_ready, dma_rsp_ error	Input valid/ error; output ready	1 bit each	Acknowledges write completion no earlier than the following clock. Export completion follows the final acknowledgement.

Parameter	Default	Meaning / supported values
MAX_WIDTH	256	Every integer 1..256; maximum viewport/framebuffer width.
MAX_HEIGHT	256	Every integer 1..256, independently; non-power-of-two dimensions supported.
TILE_LOG2	3	Every integer 0..4; tile side is 1 shifted left by TILE_ LOG2. Illegal configurations use diagnostic-safe widths and terminate simulation nonzero with PARAMETER ERROR.

III Behavior and corner cases

A Exact command framing

The header contains tag, packed width/height, opcode, sixteen row-major signed Q16.16 matrix entries and three xyzw vertices. Attribute draws append eighteen signed Q16.16 words: u,v,r,g,b,a per vertex. Legal 31-word commands are G1 modes0/1, depth clears4/5 and state commands1024..1027. Legal 49-word G2 draws are2, 3,6,7,258,259,262,263; G3 draws are514,515,518,519,770,771,774,775. These are exactly24 encodings. State commands use0x400..0x403, avoiding collisions with G3 draws0x302/0x303. Opcodes0x300/0x301 remain invalid31-word commands. Rejected recognized draws still consume49 words.

B Atomic configuration

Texture0x400 uses word1 dimensions, word3 base, word4 byte stride and zero words5..30. Shader0x401 uses word1 zero, word3 program and zero words4..30. Color clear0x402 uses dimensions, word3 RGBA8 and zero words4..30; it never clears depth. Export0x403 uses matching color dimensions, base, stride and zero words5..30. Memory descriptors require four-byte alignment, stride>=4*width and full-precision last-byte address<=0xffffffff. Invalid descriptors preserve state. G3 prerequisites are checked before replay: matching initialized color, configured shader and configured texture when enabled; depth configuration remains independent.

C Transform, clipping and projection

Transform accumulates four signed32 products in signed66, floors by arithmetic shift16 and saturates to signed32. Two-dimensional modes substitute z=0,w=65536. Clipping order is w>=1,x>=-w,x<=w,y>=-w,y<=w,z>=0,z<=w. Intersections use A+floor((B-A)*fa/(fa-fb)), then exact active-plane correction. Quotient offsets narrow to signed33, addition uses signed34 and storage retains low32. Screen coordinates floor((x+w)*width*256/(2*w)) and floor((w-y)*height*256/(2*w)), clipped to viewport endpoints. Reciprocal is floor(2^40/w); projected depth uses limit65535 or16777215. Division means mathematical floor, including negative inputs. [E002]

D Coverage, depth and attributes

Orientation is normalized and zero-area fans emit no pixels. Pixel centers use half-pixel offsets; boundary ownership is dy<0 or dy==0 AND dx>0. Lambda0/1=floor(edge*65536/area); lambda2=65536-lambda0-lambda1. Depth=floor(sum(lambda*z)/65536), accepted only when strictly less than stored depth. Perspective attributes= floor(sum(lambda*q*a)/sum(lambda*q)); signed96 accumulation and low32 quotient storage preserve parent representation, without saturation. G2/G3 traverse tile rows and cropped row-major tile samples. Disabling early-Z skips tile rejection, not fragment depth testing.

E Cache and exact bilinear filtering

RGBA8 packs R in bits7:0 through A in bits31:24. Signed Q16.16 u/v are texel coordinates, clamped to texture endpoints. Sequential T00,T10,T01,T11 lookups include duplicated edge samples. Four direct-mapped4x4 lines total256 bytes; index={block_y[0],block_x[0]} with full tags. Each successful miss accepts exactly16 row-major edge-clamped reads; validity commits only after all responses succeed. Per channel, filtering computes floor((sum(texel*x_weight*y_weight)+2^31)/2^32), with weights65536-fraction and fraction and no intermediate rounding. Reconfiguration invalidates every line.

F Programmable color processing

Program bits0/1/2 enable texture, select modulation and enable blending. Fields5:3,8:6,11:9,14:12 select RGB source/destination and alpha source/destination factors. Encodings0..5 mean ZERO,ONE,SRC_ALPHA,ONE_MINUS_SRC_ALPHA,DST_ALPHA,ONE_MINUS_DST_ALPHA;6/7 and bits31:15 are invalid. Quantization=floor((clamp(attribute,0,65536)*255+32768)/65536). Texture replaces color or modulates using floor((vertex*texture+127)/255). Blending=min(255,floor((src*sf+dst*df+127)/255)), using unmodified source and old destination alpha. Exact bounded shift/add normalization avoids combinational division. Separate source-only and configured shader passes generate record fields.

G Latency, storage and export

For acceptance at E0, bilinear output becomes valid at E5 and shader output at E6 in every mode. Both hold results and reject replacement on the consumption edge. A framebuffer read executes E1 and becomes valid E2; a write executes and completes post-E1. Clear completion follows the final actual write. Export visits tile_y, tile_x,local_y,local_x and emits exactly width*height words at base+y*stride+4*x, without padding. Repeated export preserves storage. RAM arrays are not reset.

H Records and errors

Parent records retain six G1 or thirteen G2 words. G3 fragments contain [2,tag,packed_xy_fan,lambda0, lambda1,lambda2,depth,u,v,r,g,b,a,filtered_texture,source,final_color]. Terminals contain [3,tag,status,pixels_written,fan_count,tiles_tested,tiles_rejected,fragments_tested,fragments_passed,fragments_failed,texture_hits, texture_misses,dma_words,0,0,0]. Coordinates pack x[7:0],y[15:8],fan[24:16]. Status0..5 denotes success, invalid, clipped, degenerate, missing/mismatched configuration and memory-response error. Texture errors invalidate cache, suppress the failed fragment's color write/record and drain compatibility; earlier color and depth effects remain. DMA errors stop requests and count only successful acknowledgements. Neither path promises rollback.

IV Verification methods and measured results

A Software-vs-Hardware numerical agreement

Method. Independent arbitrary-precision Python model versus compiled Verilator RTL at (1,1,0),(13,11,0),(13, 11,3),(19,17,4),(256,256,3), each with seeds17,89,233.

Result. All15 combinations passed, each with48 generated legacy/G2 and48 G3 draws:720 of each class, excluding directed traffic. This is simulated-RTL agreement, not on-board rendering parity.

Evidence: E029, E032, E034, E041

B Leaf arithmetic and factor coverage

Method. 356 paired filter/shader vectors per seed, including256 fully random vectors and separate36-case RGB and alpha factor matrices; compact Icarus checks.

Result. 1,068 compiled comparisons per leaf passed, including768 fully random cases per leaf, plus100 compact Icarus comparisons. Fixed-latency and stalled-result assertions passed.

Evidence: E028, E034, E041

C Framing, cache, DMA and recovery

Method. Public-pin assertions and transaction ledgers checked all24 opcodes, immediate following commands, malformed descriptors, cache reuse/conflicts, errors, reset and complete exports.

Result. Passed exact record/request comparisons, eight read-free repetitions after initial refill,16 reads per successful miss and framebuffer cases1x1,1x17,19x1,13x11,256x256. Fifteen legal parameter tuples passed; six illegal tuples explicitly rejected.

Evidence: E026, E030, E031, E033, E041

D Sensitivity to RTL defects

Method. Compiled isolated mutants changed bilinear rounding, cache tags, shader complement, depth equality, perspective weighting and DMA address.

Result. All six original cases passed; all six mutants produced designated numerical mismatches. Compilation failure or timeout was not counted as detection.

Evidence: E042

E Connected synthesizable structure

Method. Yosys hierarchy, process, memory and structural assertions at (13,11,3) and (1,1,0).

Result. Both passed with connected state/storage, zero latches, division/modulo cells, unresolved black boxes or asserted structural problems.

Evidence: E034, E043

F Portable reproduction

Method. Fresh-directory copy of declared RTL, design documents and tests; real Icarus smoke, numerical comparisons and singleton Yosys check.

Result. Passed without parent-reference or private runtime inputs. This was a compact reproduction subset, not a second full regression.

Evidence: E044

G L1 RTL timing lint

Method. Native static hazard inspection of all22 RTL files.

Result. Passed with zero combinational division/modulo, mixed-assignment, LUT clock-gating or loop-unroll hazards.

Evidence: E046

H L2 target internal timing and resources

Method. Intel Quartus Prime17.1 TimeQuest post-fit STA at10.0 ns on Arria10 GX10AX115N2F45E1SG, standalone default parameters (256,256,3). All non-clock interfaces were virtual; non-clock input and all output paths were explicitly false-pathed. The fitter reported214 virtual pins.

Result. Passed: WNS_setup = +1.362 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 115.77 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E047

I L3 physical Arria10 FPGA HIL

Method. Three consecutive physical PCIe hardware-bridge register57 connectivity probes.

Result. 3/3 passed deterministically. Receipt tier L3_BRIDGE_CONNECTIVITY_PROBE_ONLY does not establish renderer bitstream programming or on-board workload-vector execution.

Evidence: E048

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 10	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

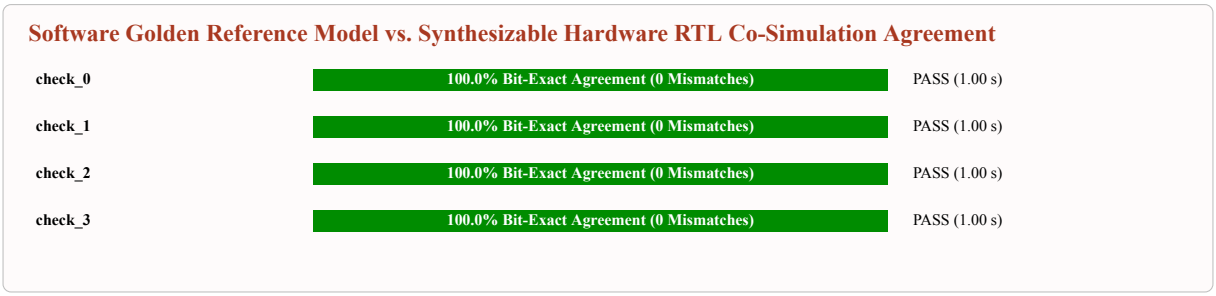


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +1.362 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

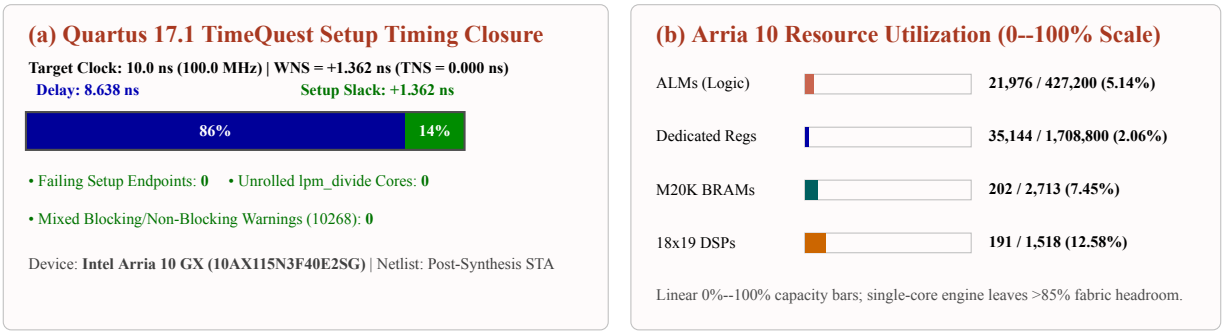


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

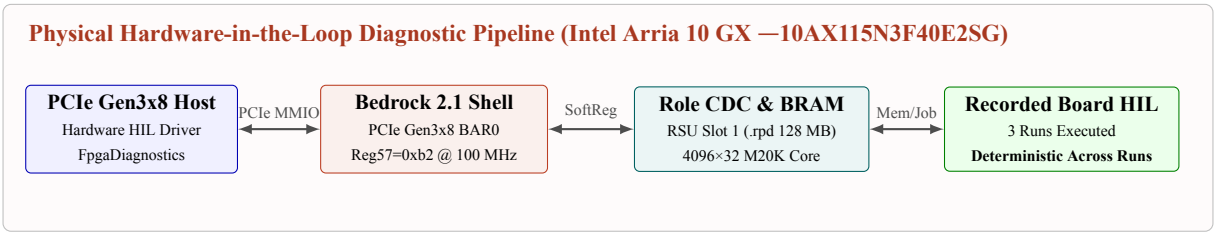


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv with crucible_g3_top as top. design/SPEC.md defines numerical/interface details; design/ARCHITECTURE.md explains implementation; tests/README.md documents verification. Copied dependencies make the reference directory unnecessary for standalone reproduction.

Reset subsystem and bridge together. After BOOT, clear depth for the intended precision, clear/configure matching color dimensions, configure the shader and configure texture if enabled. Keep texture memory immutable until reconfiguration.

Supply complete frames, maintain stalled payloads and consume terminals before issuing another command. The bridge must provide ordered eventual responses. Add integration-specific I/O, clock and bridge constraints; standalone virtual-interface STA does not constrain those paths.

With Python3, Icarus/vvp, Verilator, GNU make, g++ and Yosys installed, run python3 tests/verify.py --suite behavior, python3 tests/verify.py --suite defects, python3 tests/verify.py --suite synthesis and python3 tests/verify.py --suite reproduce. Each creates fresh source-bound evidence. Native target timing and physical gates remain separate.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 22 Verilog/SystemVerilog files; 3206 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/crucible_g1_top.sv	215 total	212 code
rtl/crucible_g2_top.sv	99 total	99 code
rtl/crucible_g3_top.sv	416 total	409 code
rtl/g1_clipper.sv	233 total	230 code
rtl/g1_divider.sv	20 total	20 code
rtl/g1_dot4.sv	62 total	59 code
rtl/g1_edge.sv	46 total	44 code
rtl/g1_project.sv	83 total	81 code
rtl/g1_raster.sv	201 total	197 code
rtl/g1_transform.sv	65 total	63 code
rtl/g2_clipper.sv	234 total	230 code
rtl/g2_divider.sv	130 total	127 code
rtl/g2_engine.sv	306 total	300 code
rtl/g2_interpolator.sv	124 total	123 code
rtl/g2_project.sv	97 total	96 code
rtl/g2_raster.sv	211 total	206 code
rtl/g2_setup.sv	87 total	87 code
rtl/g2_zbuffer.sv	132 total	131 code
rtl/g3_bilinear.sv	66 total	66 code
rtl/g3_framebuffer_dma.sv	174 total	173 code
rtl/g3_shader_pipeline.sv	107 total	105 code
rtl/g3_texture_cache.sv	150 total	148 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E004	rtl/crucible_g2_top.sv
E005	rtl/crucible_g3_top.sv
E026	tests/dma_tb.sv
E028	tests/leaf_tb.sv
E029	tests/model.py
E030	tests/parameter_tb.sv
E031	tests/reset_tb.sv
E032	tests/scenarios.py
E033	tests/top_tb.sv
E034	tests/verify.py
E041	.vulcan/engineering/run-0001/attempt-0003/stdout.log
E042	.vulcan/engineering/run-0001/attempt-0004/stdout.log
E043	.vulcan/engineering/run-0001/attempt-0005/stdout.log
E044	.vulcan/engineering/run-0001/attempt-0006/stdout.log
E046	.vulcan/engineering/run-0001/rtl-timing-lint.json
E047	.vulcan/engineering/run-0001/quartus-sta.json
E048	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).